

Mahlaka Sami

Junior Software Engineer | samimahlaka@gmail.com | 312-765-3395 | Westmont, IL | <http://www.linkedin.com/in/mahlaka-sami>

Summary

Motivated IT graduate skilled in Python, Django, Flask, and cloud technologies with experience in building scalable web apps, integrating APIs, and managing databases. Proficient in AWS, Jenkins, Linux, & translating requirements into secure, efficient solutions.

Technical Skills

- Programming Languages: Python, JavaScript
- Frameworks: Flask, Django
- Database Management: MySQL, SQLite, PostgreSQL (via pgAdmin), Redis (via RedisInsight)
- Cloud & Infrastructure: AWS (S3, EC2, IAM, Lambda), Linux
- Version Control: Git
- CI/CD Tools: Jenkins
- API Integration: Consuming and integrating APIs to retrieve and process data
- Single-Page Web Applications: Building efficient and user-friendly application
- Task Queues & Scheduling: Celery, cron

Work Experience

TechoMatrix | March 2024 – Jan 2025 |

Junior Software Engineer

- Developed and maintained web applications using Python, Django, and Flask with secure, scalable architecture.
- Created single-page web applications with Django and JavaScript fetch for seamless user interactions and real-time updates.
- Integrated APIs, managed databases, and collaborated within an Agile Scrum team.
- Collaborated within an Agile Scrum team to ensure timely delivery of features and bug fixes

Algorithm and Data Structure Proficiency

During academic period, I engaged myself in algorithm and data structure practice on platforms like **LeetCode** to enhance problem-solving abilities and coding proficiency. Developed a solid understanding of time complexity and space complexity, enabling optimization of solutions and design of efficient algorithms.

Projects

AI Chatbot App

- Developed a real-time chatbot using Django REST Framework and LLaMA, secured with JWT authentication.
- Used Redis for session handling, PostgreSQL for storing chat history, and AWS S3 for media file storage.
- Containerized with Docker and managed through Git, RedisInsight, and pgAdmin for efficient debugging and monitoring.

Polling App

- Built a polling system using Django REST with API-based voting, result fetching, pagination, and SQLite for persistent storage.
- Integrated JavaScript fetch calls for real-time vote submission and result updates without page reload.
- Added user authentication and deployed on a cloud platform to ensure high availability and smooth access.

Blog Platform

- Built a Django-based backend app integrating Cohere API for AI content creation, PostgreSQL for data storage, and Redis for session handling
 - Automated daily blog generation using cron scheduler; included mock SEO keyword analysis module.
- Focused on RESTful API design, prompt engineering, and scalable backend architecture
- Deployed on AWS with version control via GitHub and implemented secure user authentication and post management features.

Budget Tracker

- Created a Flask-based app to track personal transactions with input validation and editing features for a better UX.
- Used SQLite for data storage and Jenkins pipelines for continuous integration and deployment automation.

Academic Qualification:

2021-2023: **Master of Science (M.S.)** - Information Technology - Northeastern Illinois University, Chicago, Illinois. (3.94 CGPA)

2013-2017: **Bachelors of Software Engineering** - NED University of Engineering & Technology, Karachi, Pakistan. (Batch Position Holder – 3.97 CGPA)